

University of Southampton

Credit Card Fraud Detection using Machine Learning

COMP6237 Data Mining
Group Coursework

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The Credit Card Fraud Detection Problem

“Fraud detection is a classic anomaly detection problem where rare events must be detected within a large volume of legitimate transactions.”

- Credit card fraud causes significant financial losses globally.
- Financial institutions must detect fraudulent transactions quickly to minimise risk.
- Fraud detection is challenging because fraudulent transactions are **rare compared to legitimate ones**.
- Machine learning techniques can help identify suspicious patterns in transaction data.

Dataset Description

MLG-ULB Credit Card Fraud Dataset

(Source: Kaggle)

Key Stats

1. Total transactions: 284,807
2. Fraudulent transactions: 492
3. Fraud ratio: 0.17%

Feature Explanation

1. V1 – V28 : PCA transformed features*
2. Time : seconds elapsed between transactions
3. Amount : transaction value
4. Class : fraud label (1 = fraud, 0 = legitimate)

*PCA transformation was applied to anonymise sensitive transaction attributes.

Challenges in Fraud Detection Dataset

Extreme Class Imbalance

1. Legitimate transactions $\approx 99.83\%$
2. Fraud transactions $\approx 0.17\%$

A model predicting all transactions as legitimate would still achieve **very high accuracy**.

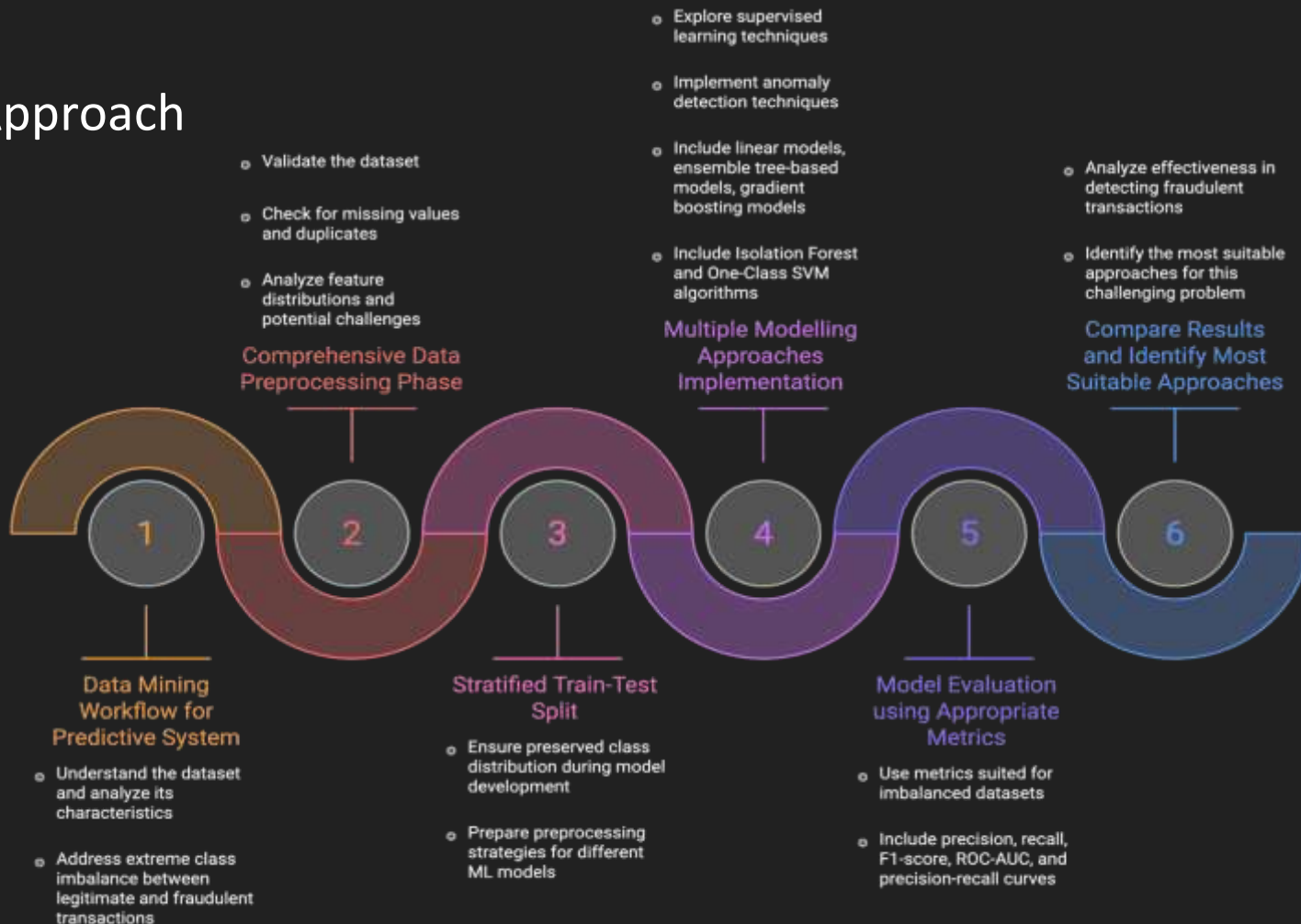
Feature Interpretability

1. Most features (V1–V28) are PCA transformed
2. Original meaning of features is unknown

PCA transformation was applied to anonymise sensitive transaction attributes.

Fraud Pattern Variability: Fraud behaviour evolves over time, making it difficult for models to generalise.

Project Approach



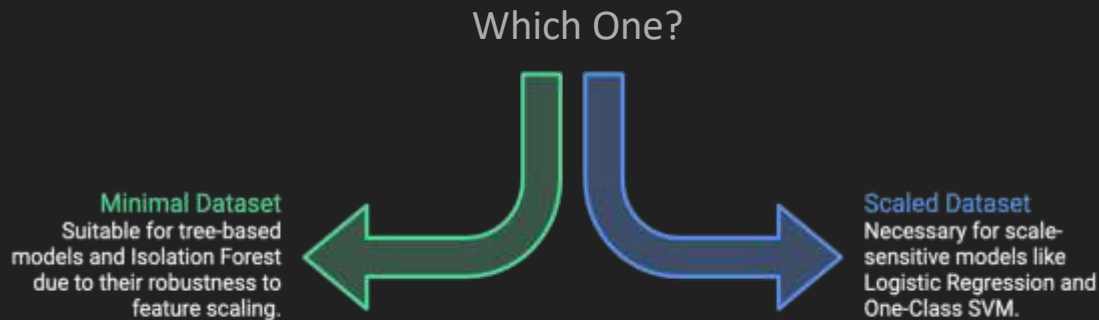
Data Preprocessing Pipeline



Model-Aware Preprocessing

At this point, we have two parallel versions of the data:

- An unscaled version for models that do not strongly depend on feature scaling.
- A robust-scaled version for models that are more sensitive to feature magnitude.



Different model families respond differently to feature scaling.

Observations after PreProcessing Stage

- The dataset was structurally clean and fully numerical with no missing values, a small number of exact duplicate rows were identified and removed to prevent repeated observations influencing the training process.
- The main challenge was the extreme class imbalance between legitimate and fraudulent transactions. Feature inspection also showed that the Amount variable was strongly right-skewed.
- To support different model types, both minimal and robust-scaled datasets were prepared, creating a preprocessing workflow that is reproducible, leakage-safe, and suitable for multiple model families in the project.

Explored Methods

1. Logistic Regression (Linear Baseline)
2. Random Forest (Ensemble Learning)
3. XGBoost (Gradient Boosting)
4. LightGBM (Gradient Boosting)
5. Isolation Forest (Anomaly Detection)
6. One Class SVM (Anomaly Detection)

Evaluation Metrics

- Precision
- Recall
- F1-Score
- ROC-AUC
- PR-AUC

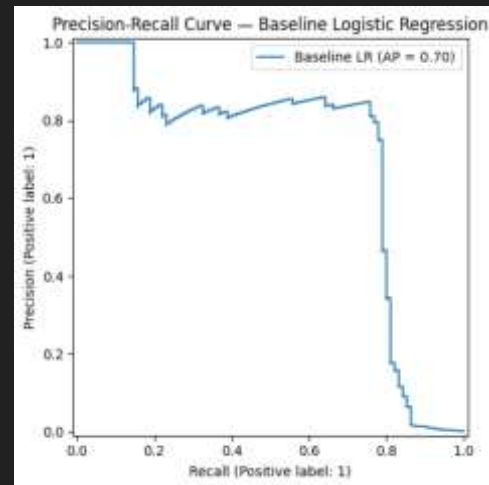
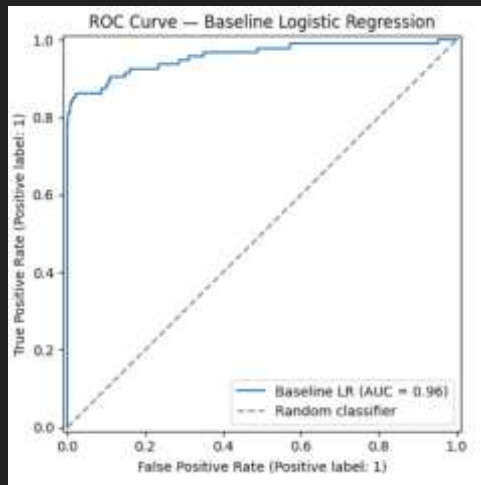
Important Note: Accuracy alone is not reliable for highly imbalanced datasets.

Logistic Regression (Baseline)

- Predicted fraud cases: 65/56,746
- Classification performance:

Classes	Precision	Recall	F1	Support
Legitimate	1.00	1.00	1.00	56651
Fraud	0.85	0.58	0.69	95

	Predicted Legitimate	Predicted Fraud
Actual Legitimate	56,641	10
Actual Fraud	40	55



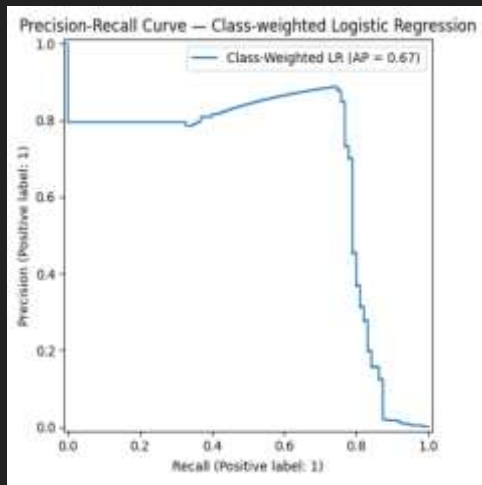
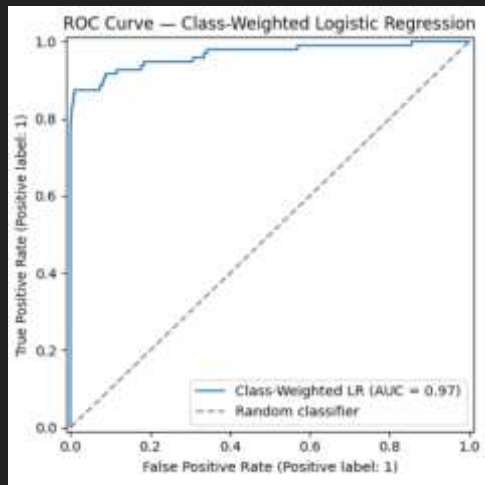
- Key Metrics:
 - ROC-AUC: 0.9581
 - PR-AUC: 0.698

Logistic Regression (Class-weighted)

- Predicted fraud cases: 1472/56,746
- Classification performance:

Classes	Precision	Recall	F1	Support
Legitimate	1.00	0.98	0.99	56651
Fraud	0.06	0.87	0.11	95

	Predicted Legitimate	Predicted Fraud
Actual Legitimate	55,262	1389
Actual Fraud	12	83



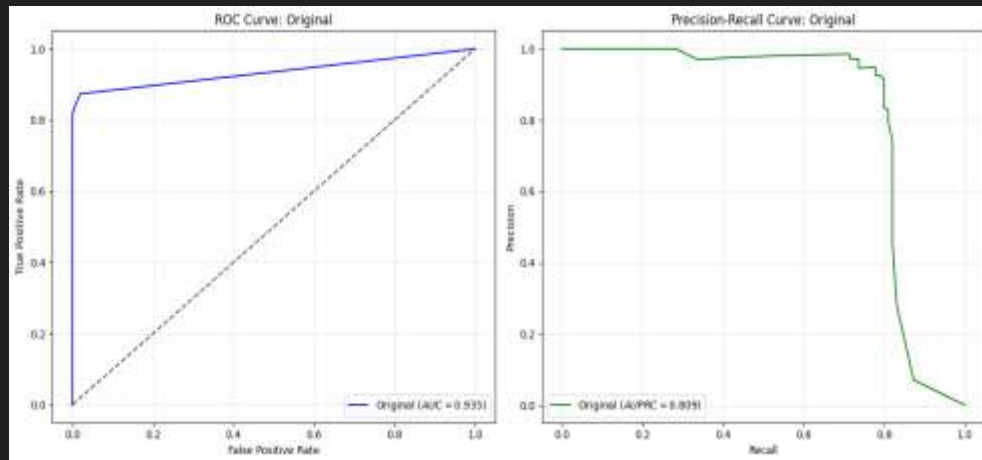
- Key Metrics:
 - ROC-AUC: 0.9657
 - PR-AUC: 0.6718

Random Forest (Original)

- Predicted fraud cases: 69/56,746
- Classification performance:

Classes	Precision	Recall	F1	Support
Legitimate	1.00	1.00	1.00	56651
Fraud	0.99	0.72	0.83	95

	Predicted Legitimate	Predicted Fraud
Actual Legitimate	56,650	1
Actual Fraud	27	68



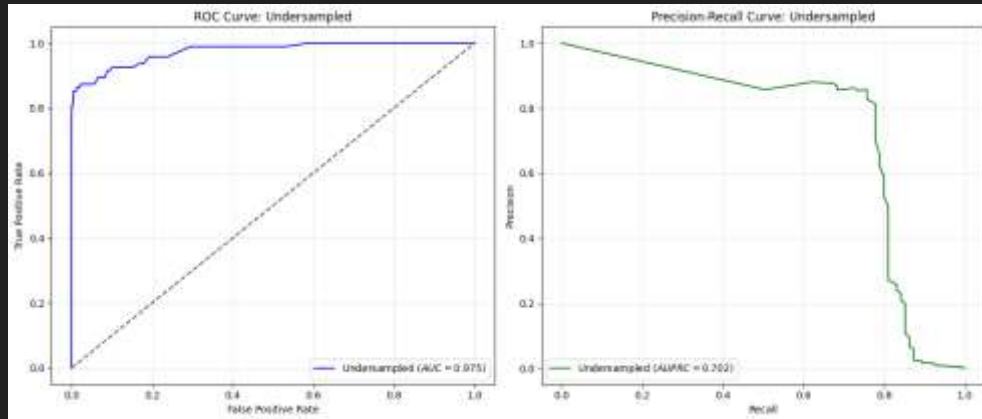
- Key Metrics:
 - ROC-AUC: 0.935
 - PR-AUC: 0.809

Random Forest (Undersampled)

- Predicted fraud cases: 1,025/56,746
- Classification performance:

Classes	Precision	Recall	F1	Support
Legitimate	1.00	0.98	0.99	56651
Fraud	0.08	0.86	0.15	95

	Predicted Legitimate	Predicted Fraud
Actual Legitimate	55,708	943
Actual Fraud	13	82



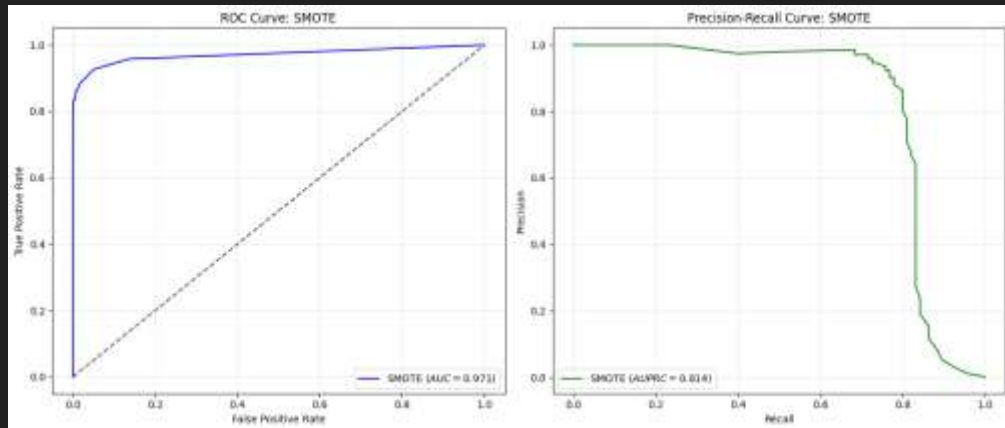
- Key Metrics:
 - ROC-AUC: 0.975
 - PR-AUC: 0.702

Random Forest (SMOTE)

- Predicted fraud cases: 78/56,746
- Classification performance:

Classes	Precision	Recall	F1	Support
Legitimate	1.00	1.00	1.00	56651
Fraud	0.92	0.76	0.83	95

	Predicted Legitimate	Predicted Fraud
Actual Legitimate	56,645	6
Actual Fraud	23	72



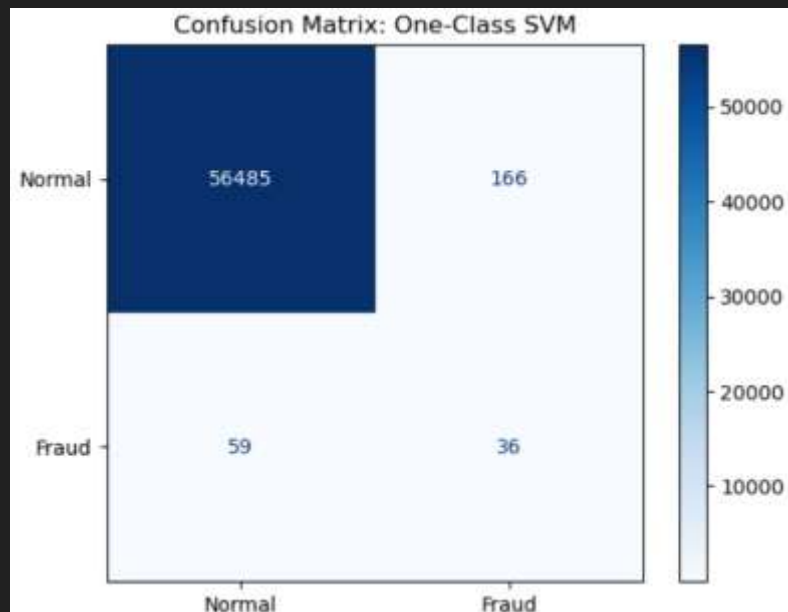
- Key Metrics:
 - ROC-AUC: 0.971
 - PR-AUC: 0.814

One-Class SVM (Unsupervised Anomaly Detection)

- Detected Fraud Cases - 44/95
- The model detected **46% of fraud cases** but produced false positives, which is typical for anomaly detection models.
- Classification Performance:

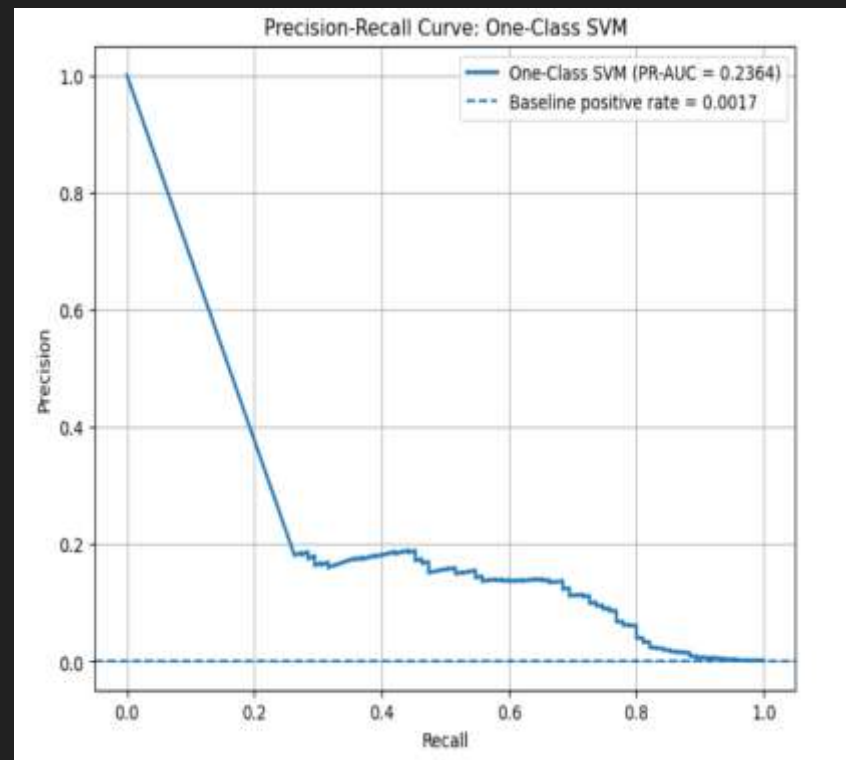
Class	Precision	Recall	F1-Score	Support
0	0.9990	0.9971	0.9980	56651
1	0.1782	0.3789	0.2424	95

- Confusion Matrix



One-Class SVM Model Analysis

- **PR-AUC is used as the primary evaluation metric** because the dataset is highly imbalanced (only ~0.17% fraud transactions).
- The **baseline precision is 0.0017**, representing the performance of random guessing.
- The One-Class SVM achieved a **PR-AUC of 0.2364**, significantly outperforming the baseline.
- The curve illustrates the **tradeoff between precision and recall**, showing the model's ability to rank fraudulent transactions higher than legitimate ones.



XGBoost — Credit Card Fraud Detection | End-to-End Pipeline

Role B1 • 284,807 transactions • 0.17% fraud rate

Dataset

284,807 transactions
492 fraud (0.17%)
31 columns → 30 features
V1-V28: PCA components
+ scaled_amount, scaled_time

Preprocessing

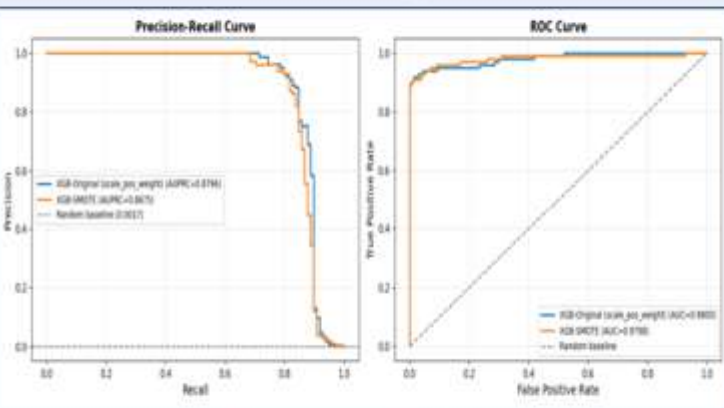
StandardScaler on Amount & Time
Drop original Amount / Time
Stratified 80/20 split
Train: 227,845 rows
Test: 56,962 rows
5-fold stratified CV indices

Imbalance Strategy

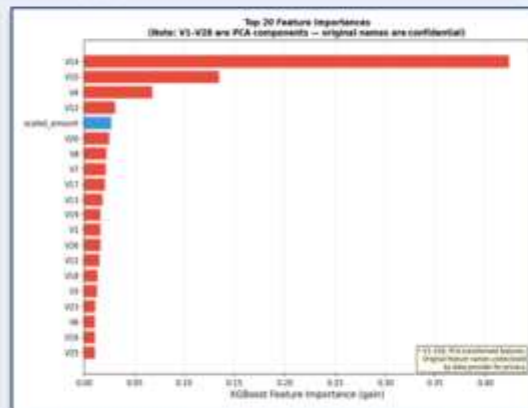
Model A — Original data
scale_pos_weight = 577
(neg + pos ratio)
Model B — SMOTE
Synthetic fraud samples
50 / 50 balanced (454,902)

Best Hyperparameters

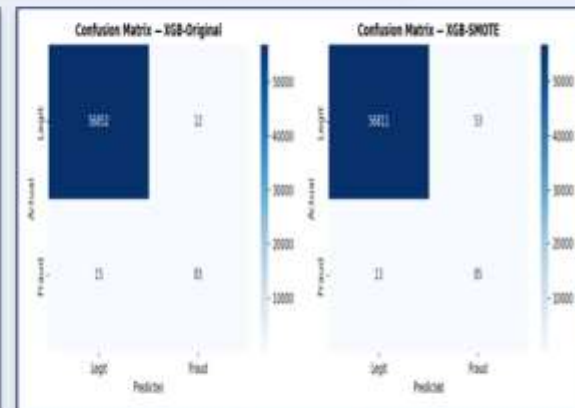
n_estimators : 400
learning_rate : 0.05
max_depth : 6
subsample : 0.8
colsample_bytree: 0.8
CV AUPRC : 0.8534



Precision-Recall & ROC Curves



Feature Importance (V1-V28 = PCA components, names confidential)



Confusion Matrices

Model	ROC-AUC	AUPRC	Fraud Recall	Fraud Precision	Accuracy
XGB — Original (scale_pos_weight)	0.9800	0.8796	0.847	0.874	99.95%
XGB — SMOTE (balanced 50/50)	0.9798	0.8675	0.867	0.616	99.88%

★ Best model on AUPRC | Grid search: 32 combos x 5-fold CV | Metric: AUPRC (fraud-focused)

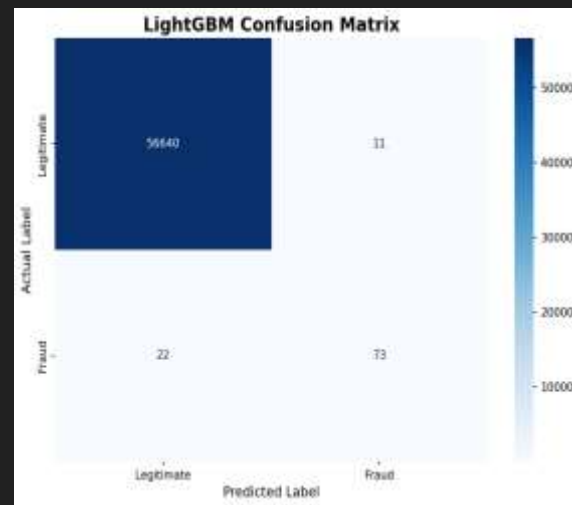
VERDICT

- ✓ XGB-Original wins on AUPRC (0.8796)
- ✓ Scale_pos_weight beats SMOTE in precision (0.874 vs 0.616)
- ✓ SMOTE: higher recall but 3x more false positives — costly for banks
- ✓ Both models: ROC-AUC ≈ 0.98

LightGBM - Model Results

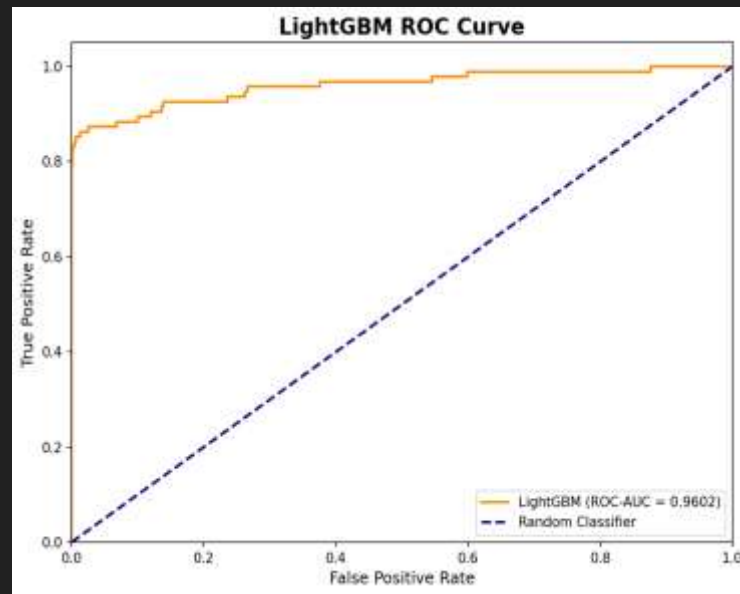
- **Scale_pos_weight = 599.48** applied to handle class imbalance
- **Predicted fraud cases: 73/95** detected
- **73 fraud cases correctly identified** out of 95
- **Only 11 legitimate transactions** incorrectly flagged as fraud
- **22 fraud cases missed** by the model

Metric	Score
Precision	0.87
Recall	0.77
F1 Score	0.82
ROC-AUC	0.9602
PR-AUC	0.8041



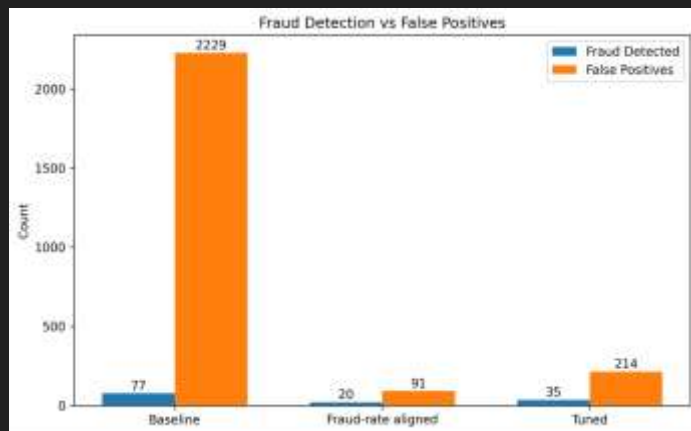
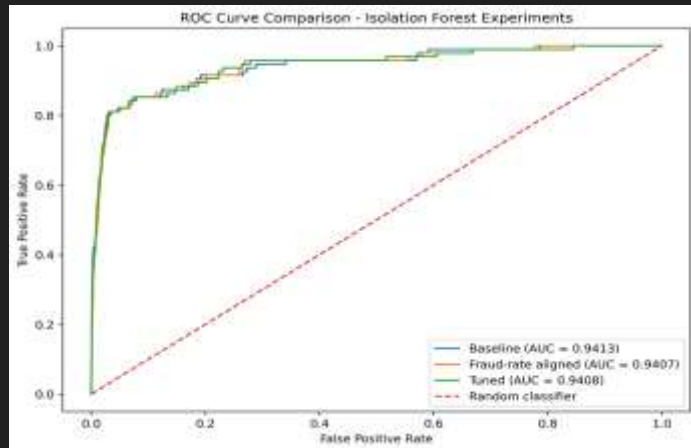
LightGBM - Model Analysis

- ROC-AUC of 0.9602 confirms strong ability to separate fraud from legitimate transactions
- PR-AUC of 0.8041 demonstrates robust performance on heavily imbalanced data
- High precision of 0.87 means very few false alarms raised
- Recall of 0.77 ensures most fraudulent transactions are successfully detected
- LightGBM outperforms anomaly detection models while remaining competitive with other supervised methods



Experimentations with Isolation Forest

- Three Isolation Forest configurations were evaluated to understand how parameter settings influence fraud detection performance.
- Aligning the contamination parameter with the dataset's fraud rate significantly reduced false positives but also decreased the number of detected fraud cases.
- A tuned configuration provided a compromise between these behaviours, detecting **35 fraud cases** while maintaining a substantially lower number of false positives compared to the baseline model.



Experiment	True Negatives	False Positives	False Negatives	True Positives	Fraud Detected	Fraud Missed	Predicted Anomalies	Actual Fraud Cases	Precision	Recall	F1 Score	ROC-AUC
Baseline	54422	2229	18	77	77	18	2306	95	0.033391	0.810526	0.064140	0.885590
Fraud-rate aligned	56560	91	75	20	20	75	111	95	0.180180	0.210526	0.194175	0.804460
Tuned	56437	214	60	35	35	60	249	95	0.140562	0.368421	0.203488	0.682322

Key Learnings

- Isolation Forest is capable of identifying anomalous transactions without relying on labelled fraud patterns.
- However, the experiments show that the model is highly sensitive to parameter configuration, particularly the contamination parameter.
- Anomaly detection models may produce higher false positives because they flag unusual patterns rather than optimising classification boundaries.
- The results show a clear trade-off between fraud detection capability and the number of false alarms, highlighting the importance of careful parameter tuning and appropriate evaluation metrics when applying anomaly detection to highly imbalanced fraud datasets.
- Because the dataset is highly imbalanced, we evaluated model performance using multiple metrics and visual comparisons rather than relying on a single metric.

Challenges and Limitations with Isolation Forest

- The contamination parameter strongly influences model behaviour, requiring careful experimentation to balance fraud detection performance with false positive rates.
- And limitation is that anomaly detection algorithms do not directly optimise classification boundaries, which can result in lower recall compared to supervised machine learning models.
- Reducing false positives while maintaining strong fraud detection is therefore a key challenge when applying anomaly detection methods.

Conclusion

- At this stage of the project, each team member has implemented and experimented with their assigned machine learning model using the shared preprocessing pipeline. Initial experiments and evaluation results have been generated for all models, including both supervised learning approaches and anomaly detection techniques.
- The next step in the project is to conduct a collaborative review session as a team to discuss the results obtained from each model. During this discussion, we will compare performance metrics, analyse the strengths and limitations of each approach, and evaluate how different models handle the challenges of the highly imbalanced fraud detection dataset.
- Based on the outcomes of this discussion, we will determine the most promising modelling strategies and identify potential improvements such as parameter tuning, evaluation refinements, and further experimentation. These insights will guide the final phase of the project, where we will refine our models, perform deeper comparative analysis, and prepare the final report.

Questions?